

Executive Summary

BACKTRACK: Your GPS for Learning

University of the Philippines Manila

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Being told to review does not tell a learner where to start. A short study break can disappear into searching broad lessons when the obstacle is one earlier skill. BACKTRACK uses the current task and the learner's answer to find a useful starting check, offer a focused Khan repair, and test the return.

A learner or teacher selects a destination. Short mathematics checks investigate relevant prerequisites. Demonstrated knowledge removes review from the route; difficulty can introduce an earlier step. The learning stop presents a short explanation, a worked example, and practice inside BACKTRACK. Official Khan videos focus on the relevant part, with an option to continue. Fresh checks then test the return. Progress stays saved when life interrupts.

The public demo includes quadratic equations, equations with brackets, adding fractions, ratios, and reading linear graphs. These sample routes show a recovery workflow that can extend across reviewed curriculum destinations, for schools and self-study.

For November 2026 through March 2027, we propose 80-120 learners across two or three school cohorts, subject to agreements, permissions, and access. Teachers will confirm destinations, assign relevant Khan practice, and supervise separate return assessments. Our signature outcome is retained reentry: initially blocked learners who pass a fresh destination task and still pass a parallel task 7-14 days later.

With UP Manila's endorsement, our team will measure learning, repeated relevant Khan use, teacher effort, and actual costs. The pilot will leave another educator with a practical kit for running and measuring the same recovery routine.